Tutorial 12

MAT3007 Optimization

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LP relaxation

- In general, IP is very hard to solve. We can relax all integral or binary constraints and solve it as a linear program.
- There are some cases in which solving the LP relaxation can give one very good solution.
- If the optimal solution to the LP relaxation is integral, then the solution must be optimal to the IP problem (Theorem).

Special LP relaxation: Totally Unimodularity

- Totally Unimodularity: A matrix A is said to be totally unimodular if the determinant of each submatrix of A is either 0, 1, or 1.
- If the constraint matrix A is totally unimodular and b is integral, then all the BFS are integers and the LP must have an optimal solution that is an integral solution.

Totally Unimodularity Example1

Recall the matching problem:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^{n} \sum_{j=1}^{n} v_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^{n} x_{ij} = 1 \qquad \forall j \\ & \sum_{j=1}^{n} x_{ij} = 1 \qquad \forall i \\ & x_{ij} \in \{0, 1\} \end{aligned}$$

The constraint matrix is TU.
An example when $n = 2$ (in the order of $x_{11}, x_{12}, x_{21}, x_{22}$):

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

One can choose $B = \{1, 2\}$, $C = \{3, 4\}$. The condition holds and A is TU.

Totally Unimodularity Example2

Matrices that are not TUM:

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Matrices that are TUM:

$$\begin{pmatrix} 1 & -1 & 0 & 0 & -1 \\ -1 & 1 & -1 & 0 & 0 \\ 0 & -1 & 1 & -1 & 0 \\ 0 & 0 & -1 & 1 & -1 \\ -1 & 0 & 0 & -1 & 1 \end{pmatrix} \qquad \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & -1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Totally Unimodularity Example3

Consider the following matrix A:

$$A = \begin{bmatrix} -1 & -1 & 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{bmatrix}$$

Then we can make the partition to be $B = \{1, 2, 3, 4\}$, $C = \phi$. The condition holds and A is TU

- If the constraint matrix A is totally unimodular and b is integral, then all the BFS are integers and the LP must have an optimal solution that is an integral solution.
- However, TU property is uncommon in practice - if one solves the LP relaxation and find the optimal BFS is not integral, it means that TU doesn't hold

General Idea of Branch-and-Bound

- Divide the feasible region into several non-overlapping parts, solve the optimization problem on each part, compare the solution in each part, and find the optimal solution to the original problem.

General Idea of Branch-and-Bound

First, solve the LP relaxation of the IP:

- If the solution is already integral, then it is optimal to the IP

If the optimal solution to the LP relaxation is $\boldsymbol{x}^*$ and x_i^* is fractional, then branch the problem into the following two:

1. One with an added constraint that $x_i \leq \lfloor x_i^* \rfloor$, call this (S1)
2. One with an added constraint that $x_i \geq \lceil x_i^* \rceil$, call this (S2)

Get two IP's after branching.

- Then solve (S1) and (S2) and assume we can get the optimal solutions $\boldsymbol{x}_1^*$ and $\boldsymbol{x}_2^*$ with optimal values v_1^* and v_2^* (both (S1) and (S2) are still integer programs).

Claim

- *If $v_2^* \leq v_1^*$, then $\boldsymbol{x}_1^*$ is the optimal solution to the original IP*
- *If $v_1^* \leq v_2^*$, then $\boldsymbol{x}_2^*$ is the optimal solution to the original IP*

The claim is true because the union of the feasible regions in each branch equals the feasible region of the original problem

Complexity of Branch-and-Bound

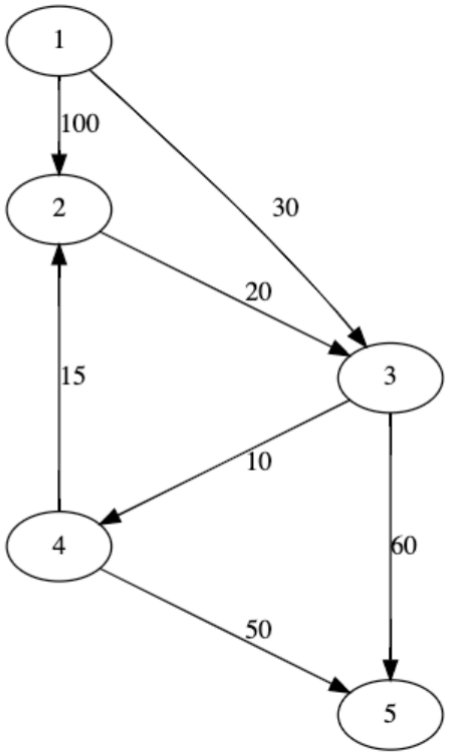
- In the worst case, branch-and-bound may need to go through each feasible integer solutions in the region, which is exponential in the problem size.
- However, branch-and-bound does enumeration in a smart way and typically it only needs to visit a tiny fraction of all solutions

Exercise 1: Travelling salesman problem(TSP)

- TSP is an NP-hard problem in combinatorial optimization, important in theoretical computer science and operations research.
- Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?
- Decision variable: $X_{ij} = 1$ if the path goes from city i to city j, $X_{ij} = 0$ otherwise.
- Take $c_{ij} > 0$ to be the distance from city i to city j. Please write the integer linear programming for TSP.

Exercise 2: Shortest path problem(SP)

- In graph theory, the shortest path problem is the problem of finding a path between two vertices in a graph such that the sum of the weights of its constituent edges is minimized.
- Decision variable: x_{ij} is an indicator variable for whether edge (i, j) is part of the shortest path: 1 when it is, and 0 if it is not. Take $a_{ij} > 0$ to be the cost from node i to node j.



Exercise 3: Branch-and-Bound

Solving the following problem by Branch-and-Bound algorithm.

$$\begin{aligned} &\text{Max}_x \ 4x_1 + 9x_2 + 6x_3 \\ &\text{subject to } 5x_1 + 8x_2 + 6x_3 \leq 12 \\ &x_1, x_2, x_3 \in \{0, 1\} \end{aligned}$$

Acknowledgement

Thank you for your time!